

How Accurate Are Learned World Models?

An Empirical Analysis of DreamerV3's Imagination on Snake

Josh Manchester

Department of Computer Science
University of Colorado Colorado Springs

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Model-Based RL: Learning by Dreaming

- Model-free RL learns by trial and error in the real environment
- Model-based RL learns **how the environment works** first, then plans
- DreamerV3 takes this further—it trains its policy **entirely inside the learned world model**
 - The agent “dreams” about what might happen
 - It never touches the real environment during policy optimization
- One algorithm, fixed hyperparameters, masters Atari, robotics, and Minecraft

Hafner et al., *Nature*, 2025

DreamerV3 trains inside imagined trajectories.

But how accurate are those imaginations?

If the world model predicts the wrong outcome for an action,
the agent is optimizing for a scenario that does not exist.

Does the agent need accurate dreams to play well?

Why This Matters

- Most DreamerV3 research focuses on **end-task performance**—scores, win rates, learning speed
- Nobody has systematically measured **how faithful the imagination is** to reality
- Understanding imagination accuracy could reveal:
 - Why model-based RL works despite imperfect models
 - What the world model actually learns to represent
 - Where model-based RL breaks down

Why Snake?

- **Simple, deterministic dynamics**
 - Snake moves one cell per step
 - Collisions end the game
 - Only randomness: food placement
- **Fully interpretable state**
 - Snake head position
 - Body segment locations
 - Food location
- **Easy to measure errors**—when the model imagines wrong, I can pinpoint exactly what went wrong
- Not yet studied with DreamerV3

Snake as an MDP

$\mathcal{S}$	$N \times N$ grid
$\mathcal{A}$	{Up, Down, L, R}
R	+1 food, -1 collision
γ	0.999

DreamerV3 Architecture Overview

- **World Model** (RSSM—Recurrent State-Space Model):
 - Encodes observations into latent states
 - Deterministic path (GRU) captures persistent information
 - Stochastic path models environment uncertainty
 - Predicts next states, rewards, and reconstructed observations
- **Actor-Critic** trained entirely in imagined trajectories:
 - Actor selects actions in latent space
 - Critic estimates value of imagined states
 - No real environment interaction during policy learning
- Key innovations: symlog predictions, fixed hyperparameters across domains

Built on: Hafner et al. (2019, 2020, 2021, 2025)

Method: Training and Trajectory Extraction

Step 1: Train DreamerV3 on Snake

- Gymnasium-compatible Snake environment, 64×64 RGB observations
- 1 million environment steps, checkpoints every 50K steps (20 snapshots)

Step 2: Extract Matched Trajectories at Each Checkpoint

- 1 Record a real trajectory using the current policy
- 2 Feed the initial observation into the world model
- 3 Roll out the world model with the **same action sequence**
- 4 Compare real vs. imagined frames side by side

Same actions $\rightarrow$ only difference is the world model's prediction accuracy.

Measuring Imagination Accuracy

Frame-Level Metrics

- **MSE**—pixel-level error
- **SSIM**—structural fidelity

Captures overall visual quality of the world model's predictions.

Semantic Metrics

- Head position error
- Body segment accuracy
- Food location accuracy
- Collision prediction

Captures whether the model understands the *game state*, not just the pixels.

Key insight: Blurry but correct $\neq$ sharp but wrong.
Separating these tells me what the model actually learns.

Correlation: Do Better Dreams Mean Better Play?

- Track both **imagination accuracy** and **game performance** across 20 checkpoints
- Three possible outcomes:

Outcome	What It Means
Accuracy $\uparrow$, Performance $\uparrow$	Better dreams drive better play—imagination accuracy is critical
Accuracy flat, Performance $\uparrow$	Agent succeeds despite imperfect dreams—policy is robust to model errors
Accuracy $\uparrow$, Performance flat	Model improves but policy does not benefit—something else is the bottleneck

I will also test which **semantic metrics** (food location, collision prediction) are most predictive of performance.

Date	Milestone
Feb 19	Proposal presentation and paper
Mar 1	Snake environment, DreamerV3 training
Mar 15	Imagination extraction pipeline
Mar 31	Checkpoint analysis complete
Apr 1	Midterm presentation and paper
Apr 15	Correlation and semantic metrics
May 7	Final presentation
May 12	Final paper

Hardware: RTX 5070 Ti (16GB VRAM), PyTorch 2.7.0, CUDA 12.8

Questions?

Josh Manchester
jmanches@uccs.edu